Problem Set 4 · Optimal bundles and kinked budget lines

Microeconomics (040184) · Group 1 · WS 2026/27 · **Solutions**

How this problem set works

When you use an LLM, you may let it solve a problem and then go over its solution. As you read through the solution, each step makes sense, and it is easy to think that you have understood it. Whether you have really understood it is something you only find out when you try to solve a problem on your own. This problem set is an opportunity to test this for yourself.

If this is how you approach problems, which is fine, then do so here: ask your LLM to solve Problem 1, go over its solution carefully, and make sure that you understand each of its steps. Problem 1 is optional and is not part of the quiz, so use it to learn.

Then, without using the LLM, test your understanding on Problem 2. Problem 2 is a different problem, but if you have understood Problem 1 well, you should be able to solve it. If you get stuck, do not ask your LLM right away. First go back to the solution of Problem 1 and, using the solution steps there, try to figure out how to solve Problem 2 on your own. Only if you are still stuck after giving it a good amount of time, get assistance from your LLM on Problem 2 as well. In this way, you will see how far you can get on your own, and whether what you thought you understood from the LLM's solution is actually calibrated.

An optimization question with a single budget line is an easy question. You should reach the level where it is also very easy for you, and then you should become very comfortable with questions with a kinked budget line.

Next week we will cover problems of this difficulty with different utility functions, so it is important that you become very comfortable solving problems at the level of Problem 2 on your own.

Once you get comfortable solving problems at the level of Problem 2, the next thing you need to tackle is to convert a verbal problem into equations. In the exam, I may make the question harder by not directly providing the budget sets as equations or the utility function as an equation, but by placing them in a verbal story from which you need to extract them. So you need to become proficient at doing this as well.

In the literature, one way to think of the complexity of a problem is by the difficulty of building the equational or functional form of the problem that you need to solve. Many people fail at this even though, if they were given the functional form, they could solve the rest. So it is equally important that you also master this skill of being able to write down the functional form of the problem (the utility function and the budget sets) given a textual story. You will practice this with Problem 3.

Drawing graphs will help you understand what is going on and get the general picture, but you should not rely on them to identify the optimal consumption bundle. I strongly encourage you to always draw a graph with the budget lines. Once you have found the solution, try to understand what the indifference curves must look like to give the solution you found. With practice, you should be able to picture the graph just by looking at the equations.

Problem 1 · Kinked budget lines

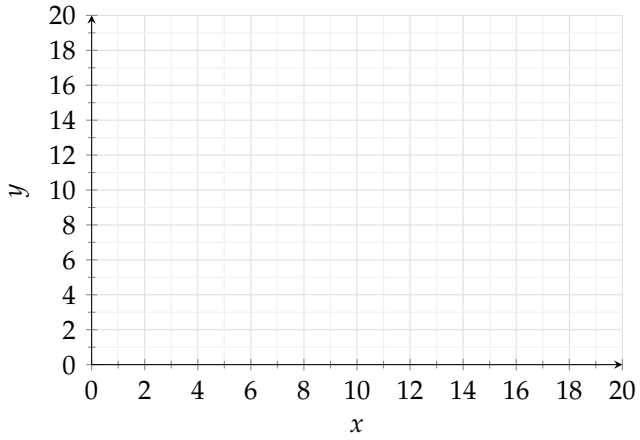
Part A · A single budget line

Notation. A **bundle** (x, y) contains x units of good x and y units of good y, and $u(x, y)$ is its **utility**. p_x and p_y are the **prices** of the two goods, and B is the **income**. The **budget set** contains all the bundles the consumer can afford: $p_x x + p_y y \leq B$. The **budget line** contains the bundles that cost exactly the income: $p_x x + p_y y = B$.

Given

A consumer has the utility function $u(x, y) = 3xy$. The prices are $p_x = p_y = €1$ and the income is $B = €10$, so the budget line is $x + y = 10$. Consider the bundles P $= (2, 8)$ and Q $= (1, 9)$.

(a) Draw the budget line in the graph below, with its end points, and mark the bundles P and Q. You will add the budget lines of Parts B and C to the same graph.



Notation. MU_x is the **marginal utility** of x: the derivative of $u(x, y)$ with respect to x, with y held fixed. MU_y is the marginal utility of y, with x held fixed. MRS is the **marginal rate of substitution**, $\text{MRS} = \frac{\text{MU}_x}{\text{MU}_y}$: the rate at which the consumer can give up y for more x and keep utility the same, as the step shrinks.

(b) At the bundle P $= (2, 8)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q $= (1, 9)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(d) The MRS is different at P and at Q. Consider these statements.

I. MU_x depends on y.
II. MU_y depends on x.
III. The prices are different at P and at Q.

Which of the statements above give the reason?

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

(e) Consider these statements about the ratio $\frac{\text{MU}_x}{p_x}$.

I. It is the "bang for the buck" on x.
II. It is the marginal utility of x per euro spent on x.
III. It is the marginal utility of x, normalized by the price of y.

Which of the statements above correctly interpret $\frac{\text{MU}_x}{p_x}$?

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

Notation. A star marks an optimal value: (x^*, y^*) denotes the optimal bundle.

(f) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part B · A change in income

Given

The utility function is still $u(x, y) = 3xy$ and both prices are still €1, but the income rises from €10 to $B =$ €18, so the budget line is now $x + y = 18$. Consider the bundles P $= (2, 16)$ and Q $= (3, 15)$.

(a) Draw the new budget line in the graph of Part A, with its end points, and mark the bundles P and Q.

(b) At the bundle P $= (2, 16)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____
(ii) The marginal utility of y: $\text{MU}_y =$ _____
(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q = $(3, 15)$, calculate the following.

(i) The marginal utility of x: MU_x = _____

(ii) The marginal utility of y: MU_y = _____

(iii) The marginal rate of substitution: MRS = _____

(d) Compare the bundle $(2, 16)$ with the bundle $(2, 8)$ from Part A. The bundle $(2, 16)$ has the same quantity of x as the bundle $(2, 8)$ and a larger quantity of y. Which of the following is correct?

- ☐ MU_x is higher at $(2, 16)$ than at $(2, 8)$, MU_y is the same at both bundles, and the MRS is higher at $(2, 16)$ than at $(2, 8)$
- ☐ MU_x is lower at $(2, 16)$ than at $(2, 8)$, MU_y is the same at both bundles, and the MRS is lower at $(2, 16)$ than at $(2, 8)$
- ☐ MU_x is the same at both bundles, MU_y is higher at $(2, 16)$ than at $(2, 8)$, and the MRS is lower at $(2, 16)$ than at $(2, 8)$
- ☐ both marginal utilities are higher at $(2, 16)$ than at $(2, 8)$, and the MRS is the same at both bundles
- ☐ both marginal utilities and the MRS are the same at both bundles

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part C · A change in price

Given

The utility function is still $u(x, y) = 3xy$ and the income is still €18, but the price of x rises from €1 to p_x = €3, so the budget line is now $3x + y = 18$. Consider the bundles P = $(1, 15)$ and Q = $(2, 12)$.

(a) Draw the new budget line in the graph of Part A, with its end points, and mark the bundles P and Q.

(b) At the bundle P = $(1, 15)$, calculate the following.

(i) The marginal utility of x: MU_x = _____

(ii) The marginal utility of y: MU_y = _____

(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q = $(2, 12)$, calculate the following.

(i) The marginal utility of x: MU_x = _____

(ii) The marginal utility of y: MU_y = _____

(iii) The marginal rate of substitution: MRS = _____

(d) The price of x is €1 in Part B and €3 in this part. Take one bundle and keep it fixed. Which of the following has a different value at this bundle when the price of x is €3 than when it is €1?

- ☐ MU_x and the MRS
- ☐ MU_x, MU_y and the MRS
- ☐ only MU_x
- ☐ only $\frac{\text{MU}_x}{p_x}$
- ☐ only the MRS

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

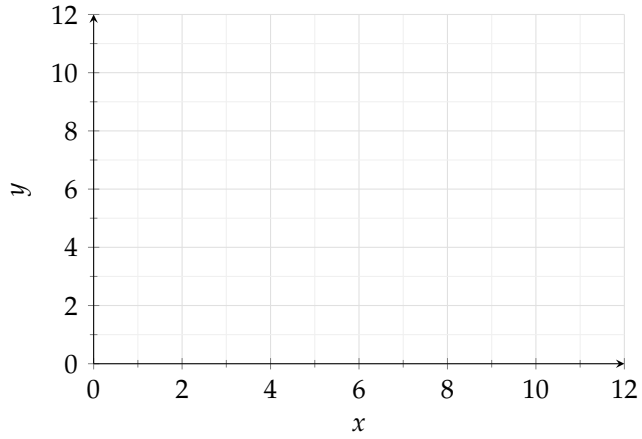
Part D · A kinked budget line

Notation. A **kink** is a point where a budget line changes slope. A **segment** is a straight part of the budget line. A **composite budget line** has one part from one budget line and one part from another budget line.

Given

The utility function is still $u(x, y) = 3xy$. The income is €10 and the price of y is €1. The first 4 units of x cost €1 each, and each further unit costs €3. The budget line therefore has two segments: $x + y = 10$ for $x \leq 4$, and $3x + y = 18$ for $x \geq 4$.

(a) Graph the composite budget line below, with its end points and its kink.



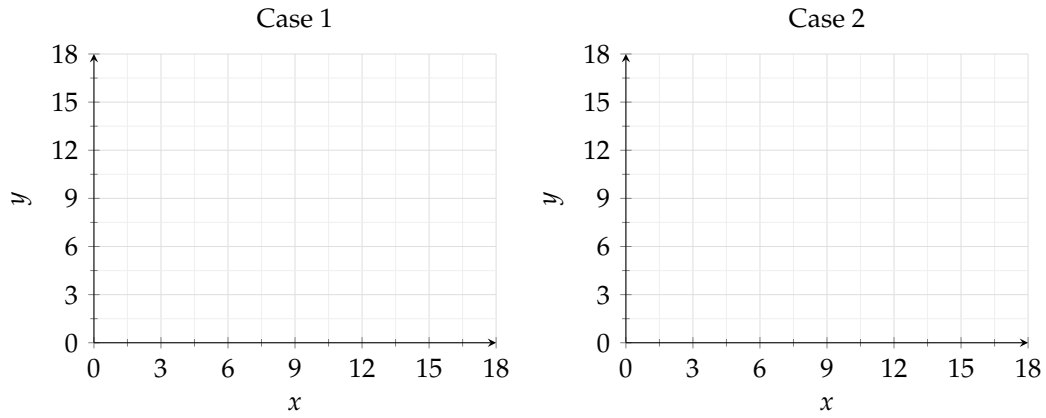
(b) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part E · Two further cases

Given

The utility function is still $u(x, y) = 3xy$ and the price of y is €1. Here are two further kinked budget lines.
Case 1. The income is €10. The first 7 units of x cost €1 each, and each further unit costs €3. The budget line is therefore $x + y = 10$ for $x \leq 7$, and $3x + y = 24$ for $x \geq 7$.
Case 2. The income is €16. The first unit of x costs €1, and each further unit costs €3. The budget line is therefore $x + y = 16$ for $x \leq 1$, and $3x + y = 18$ for $x \geq 1$.

(a) Graph the composite budget line of each case below, with its end points and its kink.



(b) Case 1. What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

(c) Case 2. What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Before Problem 2: derivatives of powers and roots

The power rule. For a power α of a variable, differentiation multiplies by α and lowers the power by 1.

$$\frac{d}{dx}\, x^{\alpha} = \alpha\, x^{\alpha-1}$$

Sums. The derivative of a sum is the sum of the derivatives. A constant factor is retained.

Negative powers. A negative power denotes a reciprocal: $x^{-\frac{1}{2}} = \dfrac{1}{\sqrt{x}}$.

A square root is a power. $\sqrt{x} = x^{\frac{1}{2}}$. So its derivative is $\frac{1}{2}\, x^{-\frac{1}{2}} = \dfrac{1}{2\sqrt{x}}$.

A constant inside the root. $\sqrt{9x} = 3\sqrt{x}$, so its derivative is $3 \cdot \dfrac{1}{2\sqrt{x}} = \dfrac{3}{2\sqrt{x}}$.

Two goods. When differentiating with respect to x, y is treated as a constant.

$$10\sqrt{xy} = 10\sqrt{y} \cdot \sqrt{x}, \quad \text{so}$$

$$\mathrm{MU}_x = 10\sqrt{y} \cdot \frac{1}{2\sqrt{x}} = 5\sqrt{\frac{y}{x}}$$

Similarly, $\mathrm{MU}_y = 5\sqrt{\dfrac{x}{y}}$. Problem 2 uses the coefficient 12 instead of 10.

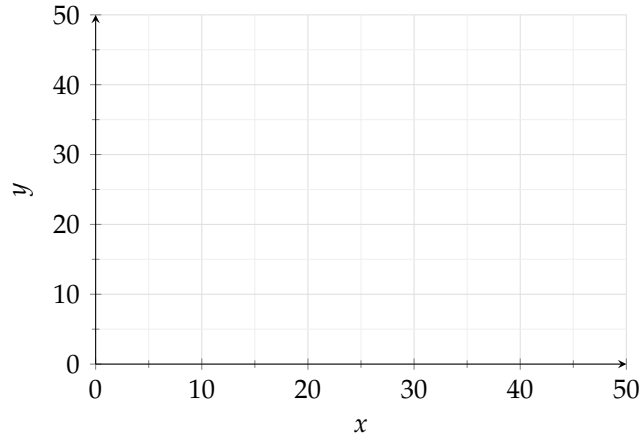
Problem 2 · Kinked budget lines and a square root

Part A · A single budget line

Given

A consumer has the utility function $u(x, y) = 12\sqrt{xy}$. The prices are $p_x = p_y = €1$ and the income is $B = €24$, so the budget line is $x + y = 24$. Consider the affordable bundles P $= (4, 16)$ and Q $= (2, 18)$.

(a) Draw the budget line in the graph below, with its end points, and mark the bundles P and Q. You will add the budget lines of Parts B and C to the same graph.



(b) At the bundle P $= (4, 16)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q $= (2, 18)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(d) The MRS is different at P and at Q. Consider these statements.

I. The ratio $\frac{y}{x}$ is different at P and at Q.

II. The coefficient 12 in the utility function is different at P and at Q.

III. Each marginal utility depends on both x and y.

Which of the statements above give the reason?

☐ only I

☐ only II

☐ I and II

☐ I and III

☐ I, II and III

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part B · A change in income

Given

The utility function is still $u(x, y) = 12\sqrt{xy}$ and both prices are still €1, but the income rises from €24 to $B =$ €48, so the budget line is now $x + y = 48$. Consider the affordable bundles P $= (8, 32)$ and Q $= (4, 36)$.

(a) Draw the new budget line in the graph of Part A, with its end points, and mark the bundles P and Q.

(b) At the bundle P $= (8, 32)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q $= (4, 36)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(d) The bundle $(4, 16)$ from Part A and the bundle $(8, 32)$ have the same MRS. What is the reason?

- ☐ the ratio $\frac{y}{x}$ is the same at both bundles
- ☐ both bundles lie on a budget line
- ☐ the income is higher in Part B than in Part A
- ☐ the prices are the same in Part A and in Part B
- ☐ the utility is the same at both bundles

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part C · A change in price

Given

The utility function is still $u(x, y) = 12\sqrt{xy}$ and the income is still €48, but the price of x rises from €1 to $p_x =$ €4, so the budget line is now $4x + y = 48$. Consider the affordable bundles P $= (1, 36)$ and Q $= (8, 8)$.

(a) Draw the new budget line in the graph of Part A, with its end points, and mark the bundles P and Q.

(b) At the bundle P $= (1, 36)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(c) At the bundle Q $= (8, 8)$, calculate the following.

(i) The marginal utility of x: $\text{MU}_x =$ _____

(ii) The marginal utility of y: $\text{MU}_y =$ _____

(iii) The marginal rate of substitution: MRS = _____

(d) The price of x is €1 in Part B and €4 in this part. Take one bundle and keep it fixed. Which of the following has a different value at this bundle when the price of x is €4 than when it is €1?

- ☐ MU_x and the MRS
- ☐ MU_x, MU_y and the MRS
- ☐ only MU_x
- ☐ only $\frac{\text{MU}_x}{p_x}$
- ☐ only the MRS

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

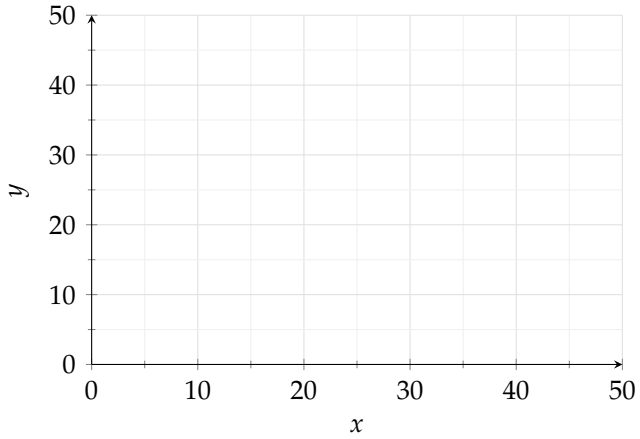
Part D · Two budget lines available

Notation. The **outer boundary** of two budget lines contains, at each x, the bundle with the highest y on either line. It is the composite budget line of the two.

Given

The utility function is still $u(x, y) = 12\sqrt{xy}$. Now the consumer can choose a bundle on either of two budget lines: $4x + y = 48$ from Part C or $x + y = 24$ from Part A. The two lines intersect, and the kink of the outer boundary, the point where they cross, lies closer to the origin than the two ends of the boundary. To the left of the kink, $4x + y = 48$ is the outer line; to the right of the kink, $x + y = 24$ is the outer line. The outer boundary is therefore $4x + y = 48$ for $x \leq 8$, and $x + y = 24$ for $x \geq 8$.

(a) Graph the composite budget line below, with its end points and its kink.



(b) Which of the following gives the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income?

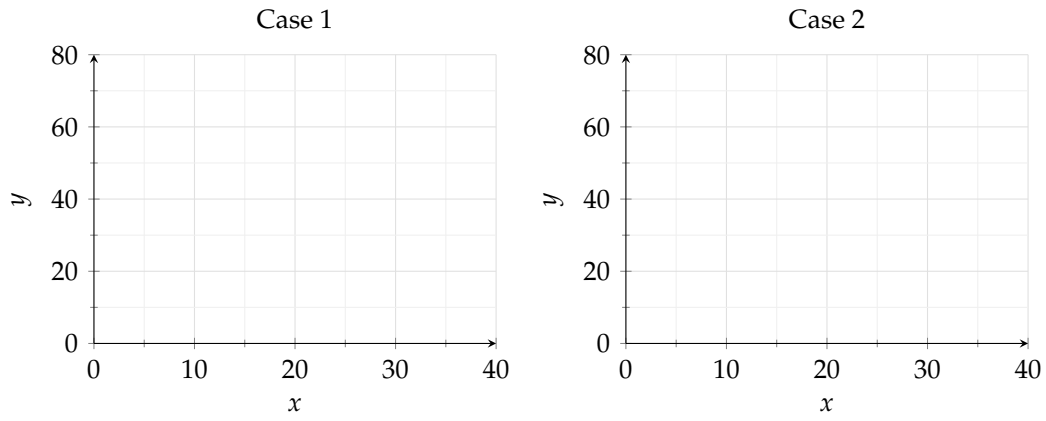
- ☐ only $(8, 16)$
- ☐ both $(4, 32)$ and $(16, 8)$
- ☐ only $(12, 12)$
- ☐ both $(6, 24)$ and $(12, 12)$
- ☐ only $(6, 24)$

Part E · Two further cases

Given

The utility function is still $u(x, y) = 12\sqrt{xy}$. In each case, the consumer can again choose a bundle on either of two budget lines.
Case 1. The two budget lines are $4x + y = 48$ and $x + y = 36$. The outer boundary is $4x + y = 48$ for $x \leq 4$, and $x + y = 36$ for $x \geq 4$.
Case 2. The two budget lines are $4x + y = 72$ and $x + y = 24$. The outer boundary is $4x + y = 72$ for $x \leq 16$, and $x + y = 24$ for $x \geq 16$.

(a) Graph the composite budget line of each case below, with its end points and its kink.



(b) Case 1. Determine the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income.

(c) Case 2. Determine the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income.

Problem 3 · From words to equations

Notation. $\min\{a, b\}$ is the smaller of the two numbers a and b.

Part A · Preferences in words

Each story below describes a consumer's preferences in words.

(a) One pack of pasta makes one meal, and the consumer is indifferent between two brands of pasta. Let x denote the number of packs of the first brand and y the number of packs of the second brand. The consumer cares only about the number of meals. Which utility function represents these preferences?

- ☐ $u = 2x + y$
- ☐ $u = \min\{x, y\}$
- ☐ $u = x + 4y$
- ☐ $u = x + y$
- ☐ $u = xy$

(b) A small pack contains 2 eggs and a large pack contains 3 eggs. The consumer cares only about the total number of eggs. Let x denote the number of small packs and y the number of large packs. Which utility function represents these preferences?

- ☐ $u = 2x + 3y$
- ☐ $u = 3x + 2y$
- ☐ $u = \min\{2x, 3y\}$
- ☐ $u = x + y$
- ☐ $u = xy$

(c) One hour of running burns 600 calories and one hour of cycling burns 400 calories. The consumer cares only about the total number of calories burned. Let x denote the hours of running and y the hours of cycling. Consider these utility functions.

I. $u = 600x + 400y$
II. $u = 3x + 2y$
III. $u = \min\{600x, 400y\}$

Which of the utility functions above represent these preferences?

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

Part B · More preferences in words

Each story below describes a consumer's preferences in words.

(a) A hot dog requires one sausage and one bun. A sausage without a bun yields no utility, and likewise a bun without a sausage. Let x denote the number of sausages and y the number of buns. The consumer cares only about the number of hot dogs. Which utility function represents these preferences?

- ☐ $u = \min\{3x, y\}$
- ☐ $u = \min\{x, 3y\}$
- ☐ $u = \min\{x, y\}$
- ☐ $u = x + y$
- ☐ $u = xy$

(b) Each sandwich requires 1 slice of bread and exactly 3 slices of cheese. Let x denote the number of slices of bread and y the number of slices of cheese. The consumer cares only about the number of sandwiches. Which utility function represents these preferences?

- ☐ $u = x + 3y$
- ☐ $u = \min\{\frac{x}{3}, y\}$
- ☐ $u = \min\{x, 3y\}$
- ☐ $u = \min\{x, \frac{y}{3}\}$
- ☐ $u = x + \frac{y}{3}$

(c) A cake requires 3 eggs and 5 spoons of sugar, always in that ratio. Let x denote the number of eggs and y the number of spoons of sugar. The consumer cares only about the number of cakes. Consider these utility functions.

I. $u = \min\{\frac{x}{3}, \frac{y}{5}\}$
II. $u = \min\{5x, 3y\}$
III. $u = \min\{3x, 5y\}$

Which of the utility functions above represent these preferences?

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

Part C · Budget lines in words

Each story below describes a consumer's budget in words. The consumer spends the entire income.

(a) A consumer has €24. The price of a coffee is €2 and the price of a sandwich is €3. Let x denote the number of coffees and y the number of sandwiches. Which equation describes the consumer's budget line?

- ☐ $2x + 3y = 12$
- ☐ $2x + 3y = 24$
- ☐ $2x + 3y = 5$
- ☐ $3x + 2y = 24$
- ☐ $x + y = 24$

(b) A consumer has €30. The first 5 gym visits cost €4 each, and each further visit costs €2. All other goods cost €1 per unit. Let x denote the number of visits and y the quantity of all other goods. Which equations describe the consumer's budget line?

- ☐ $2x + y = 30$ for $x \leq 5$, and $4x + y = 30$ for $x \geq 5$
- ☐ $4x + y = 30$ for $x \leq 4$, and $2x + y = 20$ for $x \geq 4$
- ☐ $4x + y = 30$ for $x \leq 5$, and $2x + y = 10$ for $x \geq 5$
- ☐ $4x + y = 30$ for $x \leq 5$, and $2x + y = 20$ for $x \geq 5$
- ☐ $4x + y = 30$ for $x \leq 5$, and $2x + y = 30$ for $x \geq 5$

(c) A consumer has €20. The first 3 bus rides are free, and each further ride costs €2. The consumer can take at most 10 rides. All other goods cost €1 per unit. Let x denote the number of rides and y the quantity of all other goods. Which equations describe the consumer's budget line?

- ☐ $2x + y = 20$ for $x \leq 10$
- ☐ $y = 20$ for $x \leq 3$, and $2x + y = 20$ for $3 \leq x \leq 10$
- ☐ $y = 20$ for $x \leq 3$, and $2x + y = 23$ for $3 \leq x \leq 10$
- ☐ $y = 20$ for $x \leq 3$, and $2x + y = 26$ for $3 \leq x \leq 10$
- ☐ $y = 20$ for $x \leq 3$, and $2x + y = 26$ for $x \geq 3$, with no limit on the number of rides

Solutions

The method used throughout. *Bang for the buck* compares $\frac{\mathrm{MU}_x}{p_x}$ with $\frac{\mathrm{MU}_y}{p_y}$. If the bang for the buck on x is higher, one euro moved to x raises utility, so buy more x. If it is lower, buy less x. If they are equal, stop. On a kinked budget line, use the prices of the segment you are on.

How to read the graphs. Orange is the budget line and blue is the indifference curve through the optimum. Arrows show the direction bang for the buck points. A label such as "at $(0, 10)$: 30 vs 0" gives, at that point, the bang for the buck on x against the bang for the buck on y; with several segments it starts with the segment number. A dashed circle marks a point where the two bangs for the buck are equal on the extension of a segment, off the segment itself, so that point cannot be bought.

Problem 1 · Kinked budget lines

For all of Problem 1. $u(x, y) = 3xy$. To differentiate with respect to x, treat y as a constant, and the other way round:

$$\mathrm{MU}_x = 3y, \qquad \mathrm{MU}_y = 3x$$

So

$$\mathrm{MRS} = \frac{\mathrm{MU}_x}{\mathrm{MU}_y} = \frac{3y}{3x} = \frac{y}{x}$$

Part A · A single budget line

(a) The line $x + y = 10$ ends where one good takes all the money: $(0, 10)$ and $(10, 0)$. P and Q lie on it, since $2 + 8 = 10$ and $1 + 9 = 10$. The graph is under (f).

(b) Put P $= (2, 8)$ into the formulas above.
(i) $\mathrm{MU}_x = 3 \cdot 8 = \mathbf{24}$. (ii) $\mathrm{MU}_y = 3 \cdot 2 = \mathbf{6}$. (iii) $\mathrm{MRS} = \frac{24}{6} = \mathbf{4}$.

(c) Put Q $= (1, 9)$ into the formulas.
(i) $\mathrm{MU}_x = 3 \cdot 9 = \mathbf{27}$. (ii) $\mathrm{MU}_y = 3 \cdot 1 = \mathbf{3}$. (iii) $\mathrm{MRS} = \frac{27}{3} = \mathbf{9}$.

(d) I and II.
I is true: $\mathrm{MU}_x = 3y$ depends on y, and y is 8 at P and 9 at Q.
II is true: $\mathrm{MU}_y = 3x$ depends on x, and x is 2 at P and 1 at Q.
So the two marginal utilities take different values at the two bundles, and the MRS, their ratio, is different too.
III is false: P and Q lie on the same budget line, so they face the same prices. The MRS formula $\frac{y}{x}$ contains no price at all.

(e) I and II.
I is true: this ratio is what we call the bang for the buck on x.
II is true: one unit of x costs p_x euros and adds MU_x of utility, so one euro spent on x adds $\frac{\mathrm{MU}_x}{p_x}$.
III is false: the number under MU_x is the price of x, not the price of y.

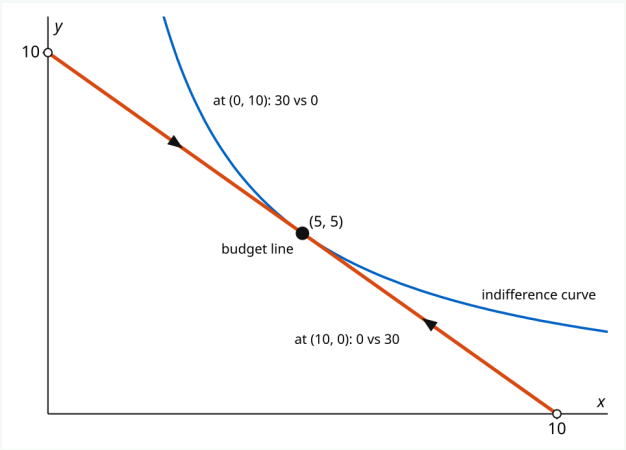
(f) Here $p_x = p_y = 1$, so the bang for the buck is $3y$ on x and $3x$ on y.
Why the optimum is where they are equal: to the left of it, the bang for the buck on x is higher, so more x raises utility. To the right of it, the bang for the buck on x is lower, so less x raises utility.

$$\frac{3y}{1} = \frac{3x}{1}, \quad \text{so } y = x$$

Put this into the budget line, because the optimal bundle must also spend all income:

$$x + x = 10, \quad \text{so } x = 5$$

Then $y = 5$. $(x^*, y^*) = (5, 5)$**,** with $u = 3 \cdot 5 \cdot 5 = 75$.



Part B · A change in income

(a) The line $x + y = 18$ runs from $(0, 18)$ to $(18, 0)$. It is parallel to the line of Part A, because the prices did not change. P and Q lie on it, since $2 + 16 = 18$ and $3 + 15 = 18$. The graph is under (e).

(b) At P $= (2, 16)$:
(i) $\text{MU}_x = 3 \cdot 16 = \mathbf{48}$. (ii) $\text{MU}_y = 3 \cdot 2 = \mathbf{6}$. (iii) $\text{MRS} = \frac{48}{6} = \mathbf{8}$.

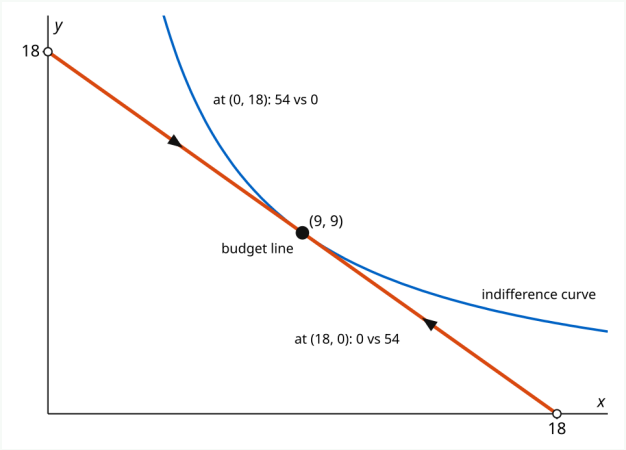
(c) At Q $= (3, 15)$:
(i) $\text{MU}_x = 3 \cdot 15 = \mathbf{45}$. (ii) $\text{MU}_y = 3 \cdot 3 = \mathbf{9}$. (iii) $\text{MRS} = \frac{45}{9} = \mathbf{5}$.

(d) MU_x **is higher at** $(2, 16)$ **than at** $(2, 8)$**,** MU_y **is the same at both bundles, and the MRS is higher at** $(2, 16)$ **than at** $(2, 8)$**.**
Compare the two bundles value by value:
MU_x is 48 at $(2, 16)$ and 24 at $(2, 8)$, because $\text{MU}_x = 3y$ and y is larger at $(2, 16)$.
MU_y is 6 at both bundles, because $\text{MU}_y = 3x$ and x is the same.
The MRS is 8 at $(2, 16)$ and 4 at $(2, 8)$.
With more y and the same x, the consumer would give up y for more x at a higher rate at $(2, 16)$ than at $(2, 8)$, keeping utility the same.

(e) The prices are still 1, so equal bang for the buck again gives $y = x$. Into the budget line:

$$x + x = 18, \quad \text{so } x = 9$$

$(x^*, y^*) = (9, 9)$, with $u = 243$. Compared with Part A, income is higher and the prices are the same, so the consumer buys more of both goods.



Part C · A change in price

(a) The line $3x + y = 18$ runs from $(0, 18)$ to $(6, 0)$. It starts at the same point on the y axis as the line of Part B, because income is the same and p_y did not change, and it is steeper, because x costs more. P and Q lie on it, since $3 + 15 = 18$ and $6 + 12 = 18$. The graph is under (e).

(b) At P $= (1, 15)$:
(i) $\text{MU}_x = 3 \cdot 15 = \mathbf{45}$. (ii) $\text{MU}_y = 3 \cdot 1 = \mathbf{3}$. (iii) $\text{MRS} = \frac{45}{3} = \mathbf{15}$.

(c) At Q $= (2, 12)$:
(i) $\text{MU}_x = 3 \cdot 12 = \mathbf{36}$. (ii) $\text{MU}_y = 3 \cdot 2 = \mathbf{6}$. (iii) $\text{MRS} = \frac{36}{6} = \mathbf{6}$.

(d) Only $\frac{\text{MU}_x}{p_x}$.
Marginal utilities come from the utility function and the bundle alone, and neither changed. So at a fixed bundle MU_x, MU_y and their ratio, the MRS, keep their values. That rules out the other four options.
The bang for the buck does contain the price. For example, at $(2, 12)$ it is $\frac{36}{1} = 36$ when $p_x = 1$, and $\frac{36}{3} = 12$ when $p_x = 3$.

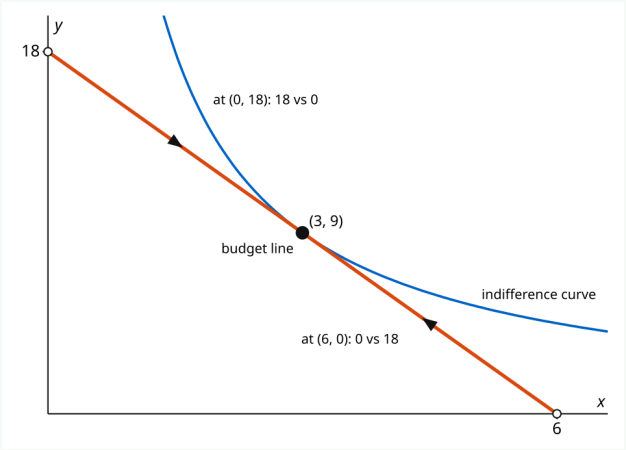
(e) Now the bang for the buck on x is $\frac{3y}{3} = y$, and on y it is $\frac{3x}{1} = 3x$. Set them equal:

$$y = 3x$$

Into the budget line:

$$3x + 3x = 18, \quad \text{so } x = 3$$

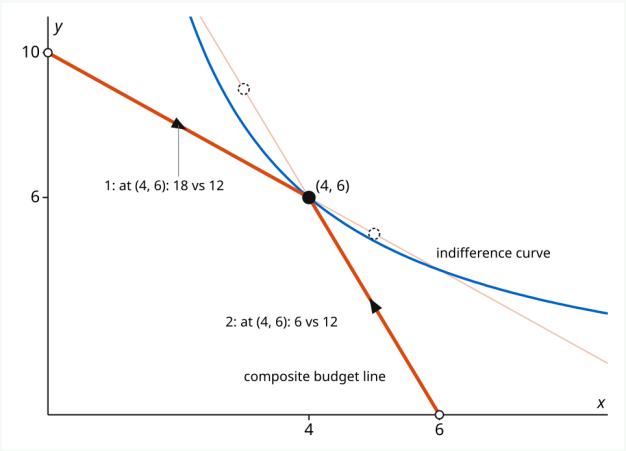
Then $y = 3 \cdot 3 = 9$. $(x^*, y^*) = (3, 9)$, with $u = 81$.



Part D · A kinked budget line

(a) The composite budget line has one part from each budget line. It starts at $(0, 10)$ and follows $x + y = 10$ to the kink at $x = 4$, where $y = 10 - 4 = 6$. From the kink $(4, 6)$ it follows $3x + y = 18$ down to $(6, 0)$. The two lines meet at the kink: $3 \cdot 4 + 6 = 18$. It is steeper after the kink because x costs more there. The graph is under (b).

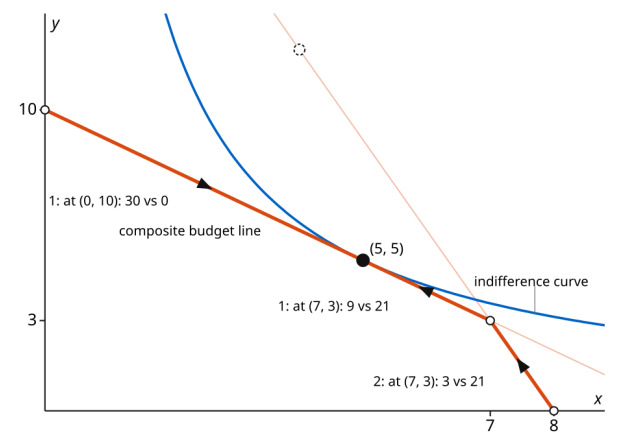
(b) Try equal bang for the buck on each segment, and keep the result only if it lies on that segment.
First segment, $x + y = 10$ for $x \leq 4$. Equal bang for the buck gives $y = x$, so $x = 5$. That is beyond $x = 4$, so it is off this segment. Everywhere on this segment the bang for the buck on x is higher, so move right, up to the kink.
Second segment, $3x + y = 18$ for $x \geq 4$. Equal bang for the buck gives $y = 3x$, so $x = 3$. That is before $x = 4$, so it is off this segment too. Everywhere on this segment the bang for the buck on x is lower, so move left, back to the kink.
Check at the kink $(4, 6)$, where $\mathrm{MU}_x = 18$ and $\mathrm{MU}_y = 12$. With the first segment's prices: 18 on x against 12 on y, so more x. With the second segment's prices: $\frac{18}{3} = 6$ on x against 12 on y, so less x. Both directions push to the kink.
$(x^*, y^*) = (4, 6)$, with $u = 72$. At this bundle the MRS is $\frac{6}{4} = \frac{3}{2}$, between the two price ratios 1 and 3, so the indifference curve touches the budget line only at the kink.



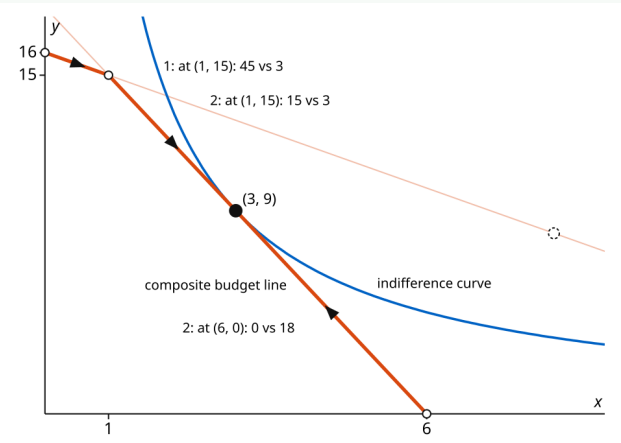
Part E · Two further cases

(a) *Case 1:* from $(0, 10)$ along $x + y = 10$ to the kink $(7, 3)$, then along $3x + y = 24$ to $(8, 0)$. *Case 2:* from $(0, 16)$ along $x + y = 16$ to the kink $(1, 15)$, then along $3x + y = 18$ to $(6, 0)$. The graphs are under (b) and (c).

(b) *Case 1, first segment* $(x \leq 7)$. Equal bang for the buck gives $y = x$, and $x + x = 10$ gives $x = 5$. That is before the kink, so it is on this segment.
Second segment $(x \geq 7)$. Equal bang for the buck gives $y = 3x$, and $3x + 3x = 24$ gives $x = 4$. That is before the kink, so it is off this segment, and bang for the buck says less x all along it. So there is one stopping point. $(x^*, y^*) = (5, 5)$, on the first segment, with $u = 75$.



(c) *Case 2, first segment* $(x \leq 1)$. Equal bang for the buck gives $y = x$, and $x + x = 16$ gives $x = 8$. That is after the kink, so it is off this segment, and bang for the buck says more x all along it.
Second segment $(x \geq 1)$. Equal bang for the buck gives $y = 3x$, and $3x + 3x = 18$ gives $x = 3$. That is after the kink, so it is on this segment.
$(x^*, y^*) = (3, 9)$, on the second segment, with $u = 81$.



The three cases of Problem 1. Part D: both points of equal bang for the buck are off their segments, so the optimum is at the kink. Part E, Case 1: the optimum is on the first segment. Part E, Case 2: the optimum is on the second segment.

Problem 2 · Kinked budget lines and a square root

For all of Problem 2. Write the utility function with powers, treating y as a constant when differentiating with respect to x:

$$u(x,y) = 12\sqrt{xy} = 12\sqrt{y}\cdot x^{\frac{1}{2}}$$

The power rule gives

$$\mathrm{MU}_x = 12\sqrt{y}\cdot\frac{1}{2\sqrt{x}} = 6\sqrt{\frac{y}{x}}$$

In the same way,

$$\mathrm{MU}_y = 6\sqrt{\frac{x}{y}}$$

The MRS is their ratio:

$$\mathrm{MRS} = \frac{6\sqrt{y/x}}{6\sqrt{x/y}} = \frac{y}{x}$$

This is the same MRS as in Problem 1, even though the marginal utilities are different.

Part A · A single budget line

(a) The line $x + y = 24$ runs from $(0, 24)$ to $(24, 0)$. P and Q are affordable and lie below the line: they cost $4 + 16 = 20$ and $2 + 18 = 20$, less than 24. The graph is under (e).

(b) At $\mathrm{P} = (4, 16)$, the ratio $\frac{y}{x}$ is 4.
(i) $\mathrm{MU}_x = 6\sqrt{4} = 6\cdot 2 = \mathbf{12}$. (ii) $\mathrm{MU}_y = 6\sqrt{\frac{1}{4}} = 6\cdot\frac{1}{2} = \mathbf{3}$. (iii) $\mathrm{MRS} = \frac{12}{3} = \mathbf{4}$.

(c) At $\mathrm{Q} = (2, 18)$, the ratio $\frac{y}{x}$ is 9.
(i) $\mathrm{MU}_x = 6\sqrt{9} = 6\cdot 3 = \mathbf{18}$. (ii) $\mathrm{MU}_y = 6\sqrt{\frac{1}{9}} = 6\cdot\frac{1}{3} = \mathbf{2}$. (iii) $\mathrm{MRS} = \frac{18}{2} = \mathbf{9}$.

(d) I and III.
I is true: $\frac{y}{x}$ is 4 at P and 9 at Q, and the MRS equals $\frac{y}{x}$.
II is false: the 12 is a fixed number in the utility function. It is the same at every bundle, and it cancels in the MRS.
III is true: $\mathrm{MU}_x = 6\sqrt{y/x}$ and $\mathrm{MU}_y = 6\sqrt{x/y}$ both contain x and y, which is why they change from bundle to bundle.

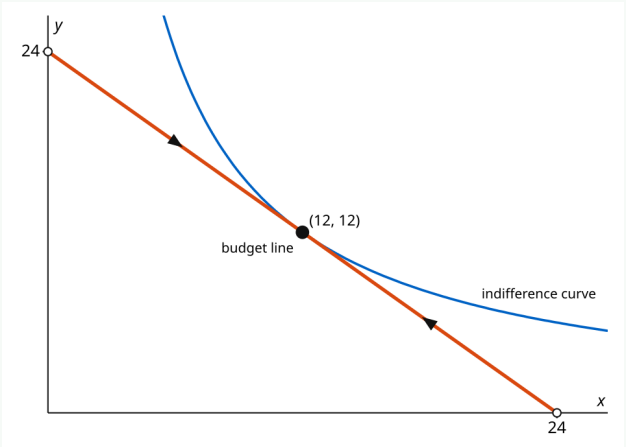
(e) With $p_x = p_y = 1$, the bang for the buck on each good is its marginal utility. Set them equal:

$$6\sqrt{\frac{y}{x}} = 6\sqrt{\frac{x}{y}}$$

Both sides are equal only when $\frac{y}{x} = \frac{x}{y}$, that is, when $y = x$. Into the budget line:

$$x + x = 24, \quad \text{so } x = 12$$

$(x^*, y^*) = (12, 12)$, with $u = 12\sqrt{144} = 144$.



Part B · A change in income

(a) The line $x + y = 48$ runs from $(0, 48)$ to $(48, 0)$, parallel to the line of Part A. P and Q are affordable: they cost $8 + 32 = 40$ and $4 + 36 = 40$, less than 48. The graph is under (e).

(b) At P $= (8, 32)$, the ratio $\frac{y}{x}$ is 4.
(i) $\text{MU}_x = 6\sqrt{4} = \mathbf{12}$. (ii) $\text{MU}_y = 6\sqrt{\frac{1}{4}} = \mathbf{3}$. (iii) $\text{MRS} = \frac{12}{3} = \mathbf{4}$.

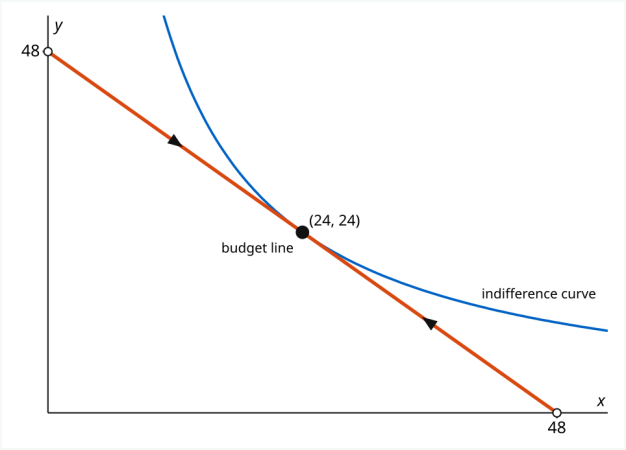
(c) At Q $= (4, 36)$, the ratio $\frac{y}{x}$ is 9.
(i) $\text{MU}_x = 6\sqrt{9} = \mathbf{18}$. (ii) $\text{MU}_y = 6\sqrt{\frac{1}{9}} = \mathbf{2}$. (iii) $\text{MRS} = \frac{18}{2} = \mathbf{9}$.

(d) The ratio $\frac{y}{x}$ is the same at both bundles. $\frac{16}{4} = \frac{32}{8} = 4$, and the MRS equals $\frac{y}{x}$.
The other options fail. Income and prices do not appear in the MRS at all, so a higher income or equal prices cannot be the reason. Lying on a budget line is not the reason either: $(4, 16)$ costs 20, so it is not on the budget line of Part A. The utility is not the same: it is $12\sqrt{64} = 96$ at $(4, 16)$ and $12\sqrt{256} = 192$ at $(8, 32)$.

(e) The prices are still 1, so equal bang for the buck again gives $y = x$. Into the budget line:

$$x + x = 48, \quad \text{so } x = 24$$

$(x^*, y^*) = (24, 24)$, with $u = 12\sqrt{576} = 288$.



Part C · A change in price

(a) The line $4x + y = 48$ runs from $(0, 48)$ to $(12, 0)$. It starts at the same point on the y axis as the line of Part B and is steeper. P and Q are affordable: they cost $4 + 36 = 40$ and $32 + 8 = 40$, less than 48. The graph is under (e).

(b) At P $= (1, 36)$, the ratio $\frac{y}{x}$ is 36.
(i) $\text{MU}_x = 6\sqrt{36} = 6 \cdot 6 = \mathbf{36}$. (ii) $\text{MU}_y = 6\sqrt{\frac{1}{36}} = 6 \cdot \frac{1}{6} = \mathbf{1}$. (iii) $\text{MRS} = \frac{36}{1} = \mathbf{36}$.

(c) At Q $= (8, 8)$, the ratio $\frac{y}{x}$ is 1.
(i) $\text{MU}_x = 6\sqrt{1} = \mathbf{6}$. (ii) $\text{MU}_y = 6\sqrt{1} = \mathbf{6}$. (iii) $\text{MRS} = \frac{6}{6} = \mathbf{1}$.

(d) Only $\frac{\text{MU}_x}{p_x}$. As in Problem 1, the marginal utilities and the MRS depend only on the utility function and the bundle. Only the bang for the buck on x contains p_x. For example, at $(8, 8)$ it is $\frac{6}{1} = 6$ when $p_x = 1$, and $\frac{6}{4} = \frac{3}{2}$ when $p_x = 4$.

(e) Equal bang for the buck now means

$$\frac{6\sqrt{y/x}}{4} = 6\sqrt{\frac{x}{y}}$$

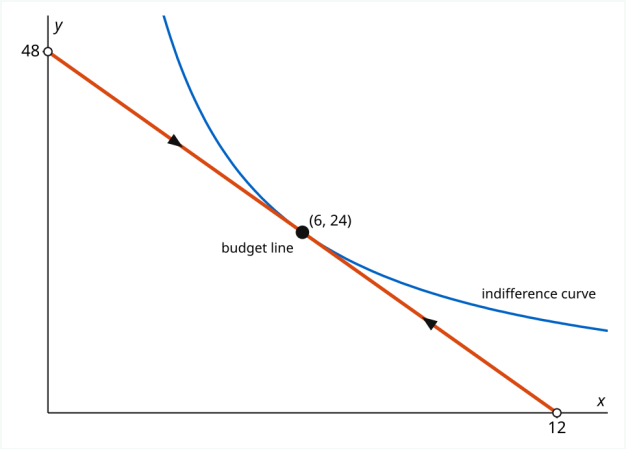
Multiply both sides by 4 and divide by $6\sqrt{x/y}$. Since $\sqrt{y/x}/\sqrt{x/y} = \frac{y}{x}$, this gives

$$\frac{y}{x} = 4, \quad \text{so } y = 4x$$

Into the budget line:

$$4x + 4x = 48, \quad \text{so } x = 6$$

Then $y = 4 \cdot 6 = 24$. $(x^*, y^*) = (6, 24)$, with $u = 12\sqrt{144} = 144$.



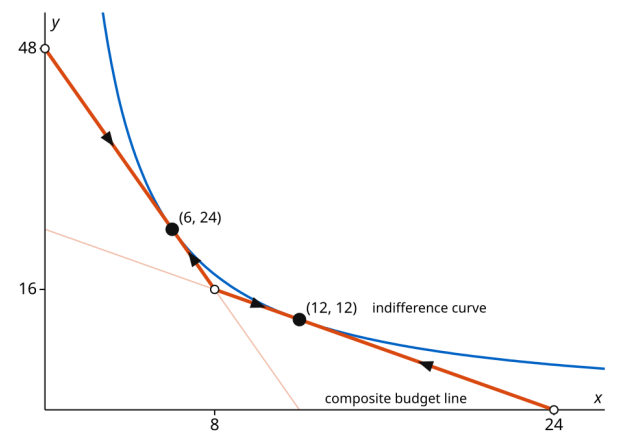
Part D · Two budget lines available

(a) The composite budget line is the outer boundary. It starts at $(0, 48)$ and follows $4x + y = 48$ to the kink $(8, 16)$, where both lines give $y = 16$. From there it follows $x + y = 24$ to $(24, 0)$. It is flatter after the kink, and the kink lies closer to the origin than the two ends. The graph is under (b).

(b) Both $(6,24)$ and $(12,12)$.
First segment, $4x + y = 48$ for $x \leq 8$. Equal bang for the buck gives $y = 4x$, as in Part C, so $(6,24)$. It lies before the kink, so it is on this segment.
Second segment, $x + y = 24$ for $x \geq 8$. Equal bang for the buck gives $y = x$, as in Part A, so $(12,12)$. It lies after the kink, so it is on this segment.
With this shape, bang for the buck stops at two points, so it alone cannot settle the answer. Compare their utilities:

$$u(6,24) = 12\sqrt{144} = 144, \qquad u(12,12) = 12\sqrt{144} = 144$$

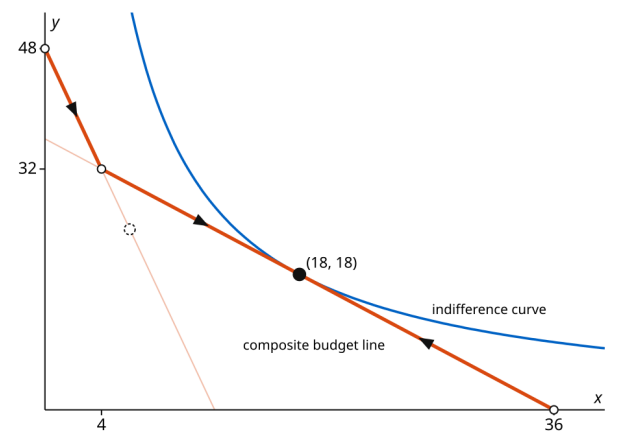
They are equally high, so both bundles are optimal.
The other options fail. At the kink, $u(8,16) = 12\sqrt{128}$, below $12\sqrt{144} = 144$, so "only $(8,16)$" is wrong; between the two optimal bundles, the kink has the lowest utility on the composite budget line. $(4,32)$ and $(16,8)$ also give $12\sqrt{128}$. "Only $(6,24)$" and "only $(12,12)$" each leave out the other bundle with the same utility.



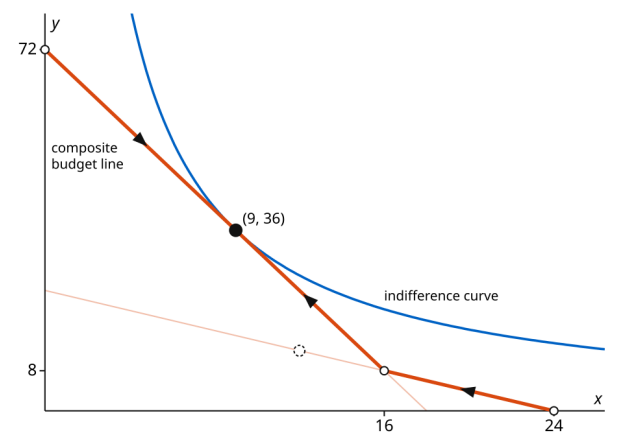
Part E · Two further cases

(a) *Case 1:* from $(0,48)$ along $4x + y = 48$ to the kink $(4,32)$, then along $x + y = 36$ to $(36,0)$. *Case 2:* from $(0,72)$ along $4x + y = 72$ to the kink $(16,8)$, then along $x + y = 24$ to $(24,0)$. The graphs are under (b) and (c).

(b) *Case 1, first segment* $(x \leq 4)$. Equal bang for the buck gives $y = 4x$, and $4x + 4x = 48$ gives $x = 6$. That is after the kink, so it is off this segment, and bang for the buck says more x all along it.
Second segment $(x \geq 4)$. Equal bang for the buck gives $y = x$, and $x + x = 36$ gives $x = 18$. That is after the kink, so it is on this segment.
$(x^*, y^*) = (18, 18)$, on the second segment, with $u = 12\sqrt{324} = 216$.



(c) *Case 2, first segment* $(x \leq 16)$. Equal bang for the buck gives $y = 4x$, and $4x + 4x = 72$ gives $x = 9$. That is before the kink, so it is on this segment.
Second segment $(x \geq 16)$. Equal bang for the buck gives $y = x$, and $x + x = 24$ gives $x = 12$. That is before the kink, so it is off this segment, and bang for the buck says less x all along it.
$(x^*, y^*) = (9, 36)$, on the first segment, with $u = 12\sqrt{324} = 216$.



The three cases of Problem 2. Part D: a point on each segment gives the same utility, so both are optimal. Part E, Case 1: the optimum is on the second segment, $x + y = 36$. Part E, Case 2: the optimum is on the first segment, $4x + y = 72$. In all three the kink is not the optimum: from the kink, moving along one of the two segments raises utility.

Problem 3 · From words to equations

How to read a story. First write down what the consumer counts in the story's own units, such as meals, eggs or hot dogs. That count is already a utility function. Any other function that ranks all bundles the same way represents the same preferences; for example, the count multiplied by a positive number.

Then ask which of two kinds the story is. If a unit of one good can replace a fixed amount of the other, the goods are **perfect substitutes**: the counts add up, and the utility is a sum. If the goods work only together, in a fixed recipe, they are **perfect complements**: the smaller count decides, and the utility is a min.

Part A · Preferences in words

(a) $u = x + y$**.**
From the words: each pack of either brand makes one meal, so the number of meals is $x + y$. The brands replace each other one for one, so they are perfect substitutes.
Why the others fail:
$u = 2x + y$ makes one pack of the first brand worth two meals.
$u = x + 4y$ makes one pack of the second brand worth four meals.
$u = \min\{x, y\}$ gives 0 with only one brand, yet 5 packs of one brand make 5 meals.
$u = xy$ also gives 0 with only one brand.

(b) $u = 2x + 3y$**.**
From the words: x small packs hold $2x$ eggs and y large packs hold $3y$ eggs, so the total is $2x + 3y$. The packs are perfect substitutes: three small packs hold the same eggs as two large packs.
Why the others fail:
$u = 3x + 2y$ swaps the two pack sizes: it gives a small pack 3 eggs.
$u = \min\{2x, 3y\}$ gives 0 with small packs alone, yet the eggs in them still count.
$u = x + y$ counts packs, not eggs.
$u = xy$ gives 0 with only one size of pack.

(c) I and II.
From the words: the calories burned are $600x + 400y$, which is I. Running and cycling are perfect substitutes for burning calories.
II is I divided by 200: $\frac{600x+400y}{200} = 3x + 2y$. Dividing by a positive number ranks all bundles the same way, so II represents the same preferences. In both, one hour of running is worth $\frac{3}{2}$ hours of cycling.
III is false: $\min\{600x, 400y\}$ gives 0 for someone who only runs, yet running alone burns calories.

Part B · More preferences in words

(a) $u = \min\{x, y\}$**.**
From the words: each hot dog needs one of each, so the number of hot dogs is limited by whichever there is less of: $\min\{x, y\}$. Sausages and buns are perfect complements. With 5 sausages and 3 buns the consumer can make 3 hot dogs.
Why the others fail:
$u = \min\{3x, y\}$ ranks $(1, 3)$ above $(1, 1)$, giving 3 and 1, yet each makes 1 hot dog.
$u = \min\{x, 3y\}$ ranks $(3, 1)$ above $(1, 1)$, giving 3 and 1, yet each makes 1 hot dog.
$u = x + y$ ranks $(4, 3)$ above $(3, 3)$, giving 7 and 6, yet each makes 3 hot dogs.
$u = xy$ does the same, giving 12 and 9.

(b) $u = \min\{x, \frac{y}{3}\}$**.**
From the words: with x slices of bread the consumer could make x sandwiches. With y slices of cheese the consumer could make $\frac{y}{3}$ sandwiches, because each takes 3 slices. The smaller number decides: $\min\{x, \frac{y}{3}\}$. Bread and cheese are perfect complements here, in the ratio 1 to 3.
Test it with the recipe: $(1, 3)$ makes one sandwich, and $\min\{1, 1\} = 1$. $(2, 6)$ makes two, and $\min\{2, 2\} = 2$. $(2, 3)$ still makes one, because there is cheese for only one, and $\min\{2, 1\} = 1$.
Why the others fail:
$u = \min\{\frac{x}{3}, y\}$ puts the 3 on the wrong good: it reads 3 slices of bread and 1 of cheese per sandwich. It ranks $(3, 1)$ above $(1, 3)$, giving 1 and $\frac{1}{3}$, yet $(3, 1)$ has cheese for only a third of a sandwich and $(1, 3)$ makes one.
$u = \min\{x, 3y\}$ is the same mistake in another form: its numbers are equal where $x = 3y$. It gives 1 at both $(1, 1)$ and $(1, 3)$, yet $(1, 1)$ has too little cheese for a sandwich.
$u = x + 3y$ and $u = x + \frac{y}{3}$ are sums: cheese alone raises them, yet cheese alone makes no sandwich.
Typical mistake: writing $3y$. The recipe says 3 slices of cheese, so the 3 goes under the cheese, as $\frac{y}{3}$.

(c) I and II.
From the words: x eggs are enough for $\frac{x}{3}$ cakes, and y spoons of sugar for $\frac{y}{5}$ cakes. The smaller number decides: $\min\{\frac{x}{3}, \frac{y}{5}\}$, which is I. It counts cakes. Eggs and sugar are perfect complements, in the ratio 3 to 5.
II is I multiplied by 15: $15 \cdot \min\{\frac{x}{3}, \frac{y}{5}\} = \min\{5x, 3y\}$. Multiplying by a positive number ranks all bundles the same way, so II represents the same preferences.
Test with the recipe $(3, 5)$*:* in I, $\frac{3}{3} = \frac{5}{5} = 1$; in II, $5 \cdot 3 = 3 \cdot 5 = 15$. In both, the two numbers in the min are equal.
III is false, and it is the tempting one: $\min\{3x, 5y\}$ copies the recipe's numbers onto the same goods. At the recipe $(3, 5)$ it compares $3 \cdot 3 = 9$ with $5 \cdot 5 = 25$, so the two numbers are not equal. They are equal at $(5, 3)$ instead: 5 eggs and 3 spoons, the wrong ratio. It also ranks $(6, 5)$ above $(3, 5)$, giving 18 and 9, yet each makes one cake.
Typical mistake: copying the recipe's numbers onto the same goods. To write a recipe of a of x and b of y without fractions, the number b goes with x and the number a goes with y: $\min\{bx, ay\}$.

Part C · Budget lines in words

(a) $2x + 3y = 24$**.**
From the words: spending on coffee is $2x$, spending on sandwiches is $3y$, and together they use all the income of 24.
Why the others fail:
$2x + 3y = 12$ uses half the income.
$2x + 3y = 5$ puts the sum of the two prices in place of the income.
$3x + 2y = 24$ swaps the prices: it charges €3 for a coffee.
$x + y = 24$ counts items, as if each cost €1.

(b) $4x + y = 30$ **for** $x \leq 5$**, and** $2x + y = 20$ **for** $x \geq 5$**.**
Up to 5 visits: each visit costs 4 and each unit of y costs 1, and spending equals 30:

$$4x + y = 30$$

At the kink: 5 visits cost $4 \cdot 5 = 20$, which leaves $y = 10$. The bundle is $(5, 10)$.
After 5 visits: each further visit costs 2, so the line has the new price and must pass through $(5, 10)$:

$$2x + y = 2 \cdot 5 + 10 = 20$$

Check: at 7 visits, the cost is $20 + 2 \cdot 2 = 24$, so $y = 6$, and $2 \cdot 7 + 6 = 20$.
Why the others fail:
"$2x + y = 30$ for $x \leq 5$, and $4x + y = 30$ for $x \geq 5$" swaps the prices.
"$4x + y = 30$ for $x \leq 4$, and $2x + y = 20$ for $x \geq 4$" has the switch at 4 visits. These two lines do not meet at $x = 4$: they give $y = 14$ and $y = 12$.
"...and $2x + y = 10$" uses the money left at the kink, 10, as the income of the second part. At $x = 5$ it gives $y = 0$, not 10.
"...and $2x + y = 30$" charges only €2 for all visits on the second part, forgetting that the first 5 cost €4. At $x = 5$ it gives $y = 20$, not 10.

(c) $y = 20$ **for** $x \leq 3$**, and** $2x + y = 26$ **for** $3 \leq x \leq 10$**.**
Up to 3 rides: rides are free, so all the money goes to other goods, and $y = 20$.
After 3 rides: each further ride costs 2, and only rides beyond the third are paid for:

$$y = 20 - 2(x - 3)$$

Multiply out and move $2x$ to the left:

$$2x + y = 26$$

Check: at $x = 3$, $y = 20$. At the limit $x = 10$, 7 rides are paid for, which costs 14, so $y = 6$, and $2 \cdot 10 + 6 = 26$.
Why the others fail:
"$2x + y = 20$ for $x \leq 10$" charges for the free rides too.
"...and $2x + y = 20$ for $3 \leq x \leq 10$", once past the third ride, charges for the 3 free rides as well: at $x = 3$ it gives $y = 14$, not 20, so the budget line jumps.
"...and $2x + y = 23$" adds 3 to the income, the number of free rides, not their value at the later price, $2 \cdot 3 = 6$. At $x = 3$ it gives $y = 17$, not 20.
"...and $2x + y = 26$ for $x \geq 3$, with no limit on the number of rides" has the right line but ignores the limit of 10 rides.